

Gold Mining Companies and the Price of Gold

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Abstract

This paper studies the exposure of Australian gold mining firms to changes in the gold price. We use a theoretical framework to formulate testable hypotheses regarding the gold exposure of gold mining firms. The empirical analysis based on all gold mining firms in the S&P/ASX All Ordinaries Gold Index for the period from January 1980 to December 2010 finds that the average gold beta is around one but varies significantly through time. The relatively low average gold beta is attributed to firm's hedging and diversification. We further find an asymmetric effect in gold betas, i.e. the gold exposure increases with positive gold price changes and decreases with negative gold price changes consistent with gold mining companies exercising real options on gold.

Keywords: gold, gold mining stocks, asymmetric payoff, real options, exchange-traded funds

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Introduction

Investors who want exposure to the price of gold can either buy gold bars, gold coins, buy shares of exchange-traded funds on gold or invest in gold mining firms. Theoretically, the latter provides a leveraged investment into gold since a share in a gold mining firm does not only give access to one unit of gold but a share in the total future production of the gold miner. If a gold mining firm further uses managerial flexibility to expand in times of rising gold prices and to contract (including the temporary closure of mines) in times of falling gold prices, gold mining shares may be a superior alternative to physical gold investments. In other words, since gold mining firms hold real options on gold with the exercise price being their marginal production costs, they can provide an asymmetric exposure to gold yielding higher returns.

This study aims to analyze the performance of gold mining firms relative to the price of gold and provide information about the value of the real options held by the firms, the hedging activities and the managerial flexibility of the firms. We focus on firms that are part of the Australian gold mining company index listed on the Australian stock exchange. Since Australia is the second largest gold producing country¹ an analysis of Australian gold mining firms can provide interesting results on its own and if compared to the literature that focussed on other countries.

The strong positive trend in the price of gold between 2004 and 2010 provides an additional motivation and feature that is particularly interesting for the question whether gold mining firms actively use their real options on gold and thus provide an asymmetric exposure to gold price changes. The 2004 - 2010 period is also characterized by the introduction and increased popularity of exchange-traded funds on gold. Since shares in exchange-traded funds are financial claims on gold and thus an alternative to both direct physical investments and equity investments an analysis of this period can provide information on how the introduction and presence of ETFs on gold has affected the exposure

¹Australia produced the second largest amount of gold (in tonnes) in 2011 and 2012. China is the largest producer and the US is the third largest producer (see Thomson Reuters GFMS Gold Survey, 2013).

of gold mining firms to the price of gold.

This study is related to earlier work that analyzes the stock price exposure of gold mining firms to gold (see Blose and Shieh, 1995, Faff and Chan, 1998, Faff and Hillier, 2004, Twite, 2002, Tufano, 1996 and Tufano, 1998). While many studies focus on North American gold mining firms (e.g. Tufano, 1996 and 1998), this research follows Faff and Chan (1998) and Twite (2002) and studies Australian gold mining firms but extends the existing literature as follows: First, we use a theoretical framework to formulate hypotheses about the gold exposure of gold mining firms. Second, by using the longest sample period among all existing studies the data comprises significant bear and bull market regimes and is thus well-suited for an analysis of asymmetric gold betas and real optionality. Relatively short sample periods may not be comprised of different gold market conditions or regimes and thus provide biased estimates of the importance of real optionality.² Third, we analyze the exposure of gold mining firms assuming constant, asymmetric and time-varying exposure.

Our estimates provide evidence for average gold betas around one. These estimates are lower than reported in Blose and Shieh (1995) and Tufano (1998) which report gold betas around 1.5 for North American gold mining companies but consistent with the findings of Coleman (2010) and Twite (2003) which report similar gold betas for Australian gold mining companies. The different gold betas estimated in this study are a result of the significantly longer sample period. Since Blose and Shieh (1995) and Tufano (1998) only use a ten-year (1981-1990) and four-year (1990-1994) sample period for their analysis, respectively, it is not surprising that the gold betas are different. A sub-sample analysis of the Blose and Shieh (1995) and Tufano (1998) sample periods yields similar gold betas. Potential economic reasons for the lower gold beta are hedging and diversification. These reasons can be clearly distinguished from the asymmetric effect of gold betas which support the hypothesis that gold mining firms exercise the real options derived from the mining activities. Finally, the significant and relatively large exposure of gold mining firms to

²The relatively stable gold price in the 1990s is an example of a period that is not comprised of different gold market conditions or regimes.

the Australian stock market illustrates that gold mining firms offer both exposure to the equity market and the gold market.

The remainder of this paper is divided into four sections. Section one contains a description of the sample, section two discusses theoretical predictions regarding the exposure of gold mining firms to the price of gold and formulates testable hypotheses, section three presents the estimation results and discusses its implications. Finally, section four summarizes the main findings and provides concluding remarks.

1 Sample Description

This section presents the sample of gold mining firms, gold prices, exchange rates and stock index data. The core of the sample consists of 41 gold mining firms that are the constituents of the ASX All Ordinaries Gold Equity Index obtained at a daily frequency from January 1, 1980 until December 31, 2010.³ Table 1 displays all gold mining firms and descriptive statistics of the weekly share price returns of each firm. The descriptive statistics are based on 30 years of weekly data between 1980 and 2010. The number of observations varies across firms and ranges between 55 and 1,617 observations. Firms with only a small number of observations are firms that became publicly listed companies only recently, e.g. Beadell Resources (in 2007), Eldorado Gold CDI (in 2009), Nobel Mineral Resources (in 2008) and Silver Lake Resources (in 2007).

The table demonstrates that the sample of firms exhibits substantial variation in the average monthly return and the risk of the companies measured by the standard deviation of the returns. Most firms entertain positive returns over the sample period but there are some firms with negative returns (e.g. Alliance Resources and Gold One International).

*** Insert Table 1 about here ***

The table also contains the descriptive statistics of the Australian stock market index

³The firms were the constituents of the index as of December 31, 2010. The data was obtained from Thomson Reuters Datastream. Two firms were eliminated since the number of observations was too low.

return and returns of gold bullion denominated in both US dollar and in Australian dollars. The table shows that the Australian stock index outperformed gold denominated in US dollars and denominated in Australian dollars. A comparison of the returns of gold mining companies with gold returns is not straightforward given the different sample periods. However, an additional analysis reveals that 21 firms outperformed gold since they are publicly listed and that the return differences converge to zero for increasing observations. This finding demonstrates that there is a disconnect between gold and gold mining shares for some firms but that more than half of the gold mining firms outperformed gold bullion consistent with the hypothesis that a gold mining company provides more leverage.

Since the gold mining firms are listed on the Australian stock exchange (ASX) it is important to analyze the role of the currency denomination on the price of gold and thus on the share prices of the gold mining companies. The evolution of the price of bullion in the two currencies is presented in Figure 1 and shows a relatively stable relationship between 1985 and 2000. From the year 2000 the Australian dollar appreciated towards parity which led to the convergence of the US dollar price and the Australian dollar price of gold.

*** Insert Figure 1 about here ***

2 Theoretical Background

The aim of this study is to assess the exposure of gold mining firms to the price of gold. Prior to the econometric specification and estimation it is important to understand the theoretical relationship between the value of a gold mining firm and its exposure to the price of gold. A direct exposure to the price of gold implies a gold beta of one. Hence, gold bullion and exchange-traded funds on gold which are designed to provide a direct exposure to gold exhibit a gold beta of one. Conversely, gold mining firms provide a leveraged investment into gold since a share in a gold mining firm provides not only

access to one unit of gold but a share of the total future production of the gold mining company. This leverage implies a gold beta that is larger than one. Blose and Shieh (1995) and Tufano (1998) both use models that demonstrate theoretically that the gold beta of an unhedged and undiversified gold mining company is larger than one. If the mining company hedges part of its gold price exposure or does not exclusively produce gold but also other commodities like copper or silver, the gold beta is reduced. A fully hedged or diversified firm would display a gold exposure of zero. Finally, if the management of the gold mining company exploits the real option character of mining it can reduce the exposure by temporarily closing mines when the price falls below the marginal cost of production and increase the exposure by exploring previously unprofitable or new mines when the price increases. This should reduce the gold beta for lower gold price regimes and increase the gold beta for higher gold price regimes. If the management uses the real options embedded in the mines it leads to an asymmetric payoff. The use of real options can thus be distinguished from hedging or diversification which lower the exposure across all gold price scenarios without an asymmetric effect.

Both Blose and Shieh (1995) and Tufano (1998) assume that the value of a gold mining company can be expressed as follows:

$$V = \sum_{i=1}^N \frac{[Q(P - C) - F]}{(1 + r)^i} \quad (1)$$

where Q is the fraction of reserves R mined per year, P is the price of gold which is assumed to be exogenous, C is the variable cost of mining and F represents the fixed costs.⁴ With a fixed production schedule and thus no flexibility the firm's (unhedged) gold beta is

$$\beta_G = \frac{\frac{\partial V}{\partial P}}{\frac{V}{P}} = \frac{PQ}{Q(P - C) - F} = \frac{P}{P - C - \frac{F}{Q}} \quad (2)$$

⁴Blose and Shieh (1995) do not explicitly integrate fixed costs into their model. Tufano (1998) integrates a corporate tax rate into his analysis and further argues that a particular fixed cost borne by mines is the fixed financial charge imposed by borrowing. Observed exposures should increase with increasing debt.

This equation replicates the main predictions formulated above. For non-zero values of C and F , the nominator is larger than the denominator yielding a gold beta that is larger than one. For the theoretical case of C and F equal to zero, a firm would be similar to gold bullion.

The above equation shows that the gold beta β_G decreases with increasing prices of gold P . Hence, the higher the price of gold *ceteris paribus*, the lower is the sensitivity of gold mining firms to variations in the price of gold. If all gold mining firms had fixed production schedules the average gold beta would be lower in a high price gold regime than in a low gold price regime.

However, empirically the fixed production schedule is a strong assumption. More realistic and common for a mining company is a flexible production model (see Brennan and Schwartz, 1985). This model explicitly integrates the real option characteristic of a gold mining firm. The manager of such a firm holds options on gold with the exercise price being their marginal production cost. The manager can increase production in high gold price regimes and decrease production in low gold price regimes. This flexibility implies a lower (smaller than one) gold beta in low gold price regimes and a higher gold beta in high price regimes. This could lead to an average gold beta of one if the two effects compensate each other or to a gold beta larger than one if the high price regime effect dominates the low price regime effect.

This section demonstrated that empirical estimates of a firm's gold beta β_G can be used to derive information about variations in the fraction of reserves mined through time, a firm's hedging activities and the degree of diversification of the firm.

To summarize, we expect betas larger than one for unhedged and undiversified firms, an asymmetric exposure for firms that use the real option characteristic of gold mines.

3 Empirical Analysis

The expectations regarding the gold betas formulated in the previous section are tested econometrically as follows. First, we estimate a model with the market return and gold

return as factors and second, we include an interaction term to identify an asymmetric effect in gold betas.

$$R_i = \alpha_i + \beta_{M,i}R_M + \beta_{G,i}R_G + \epsilon_i \quad (3)$$

where R_i , R_M and R_G are the stock returns of gold mining firm i , the market return and the return of gold, respectively. The error term is given by ϵ_i . The parameters to estimate are α_i , $\beta_{M,i}$ and $\beta_{G,i}$. The sensitivity or exposure to the stock market is captured by β_M and the sensitivity to the gold market is modeled by β_G .

We use the parameter estimates to test the hypothesis $H_0 : \beta_{G,i} = 1$.

Since it is not clear a priori which return frequency yields the most accurate gold beta, we first report summary statistics of average gold betas estimated using daily, weekly and monthly returns for different prices of gold, i.e. gold denominated in US dollars and gold denominated in Australian dollars. Table 2 shows the average market beta and gold betas across all 41 firms. The table demonstrates that both the market and the gold betas increase from daily to monthly return frequencies with market betas ranging from 0.70 for daily observations to 1.45 for monthly observations. The gold betas vary between 0.21 for daily data and 1.17 for monthly data. In addition, the table illustrates the robustness of the exposure coefficients for alternative model specifications and implicitly displays information about the correlation of equity market returns and gold returns.⁵

*** Insert Table 2 about here ***

The Table provides the estimates for five different models to allow a comparison of the isolated effect of each variable and the joint effects. The isolated and joint effects differ due to the correlation of equity index returns and gold returns.

The finding that different return frequencies yield different results is not surprising.

⁵The correlation of equity market returns and gold returns is 0.17 for gold returns denominated in US dollar and 0.06 for gold returns denominated in Australian dollar.

Since the share prices of many gold mining firms are relatively illiquid daily observations yield biased results. Moreover, it is unlikely that investors or firms react to gold price changes immediately and thus on a daily basis. It is more reasonable to assume that investors and firms use weekly or monthly gold price changes as the basis for their decisions.⁶

Table 3 presents the market and gold price (US dollar) exposures for each gold mining firm in the sample for weekly observations. The table shows that all firms display positive market and gold betas with only two exceptions. Eldorado Gold CDI exhibits a negative market beta (not significantly different from zero) but a gold beta close to one ($\beta_G = 0.927$). The other firm is Talisman Mining which exhibits a negative gold beta but a relatively large market beta ($\beta_M = 1.379$). Firms with an exposure equal to one, that is, every 1% change in the price of gold leads to a 1% change in the value of the company are Allied Gold Mining, Evolution Mining, Oceanagold CDI and Red 5. Firms that provide a larger exposure to gold price movements are Gold One International, Newcrest Mining, Resolute Mining and Silver Lake Resources.

*** Insert Table 3 about here ***

The histogram presented in Figure 2 provides a statistical summary of the estimation results for weekly returns. The histogram displays the mean and standard deviations of the market and gold betas and the distribution of those betas across all 41 firms in the sample. The average market beta is estimated at 0.9 with a standard deviation of 0.3 and the average gold beta is estimated at 0.7 with a standard deviation of 0.3. The smallest and largest gold betas are -0.01 and 1.47 , respectively. The latter estimate is particularly surprising as it implies that none of the 41 firms exhibits an exposure to gold that considerably exceeds one.

⁶Pastor and Stambaugh (1999) argue that estimates based on weekly data are likely to be more accurate in a statistical sense due to the larger number of observations but the statistical precision may be adversely affected by the higher sampling frequency and illiquidity of the stocks.

Monthly returns yield different results. The average market beta is 1.1 with a standard deviation of 0.6 and the average gold beta for monthly data is similar, i.e. 1.1 with a standard deviation of 0.6. These results imply that a 1% change in the price of gold increases the price of the average gold mining firm by 0.7% *ceteris paribus* on a weekly basis and by 1.1% on a monthly basis.

*** Insert Figure 2 about here ***

The estimates illustrate that the mean exposure of gold mining firms to the Australian stock market and the price of gold and the cross-sectional dispersion of the exposure estimates increase with decreasing return frequencies.

3.1 Asymmetric Gold Exposure

The manager of a gold mine has the flexibility to expand in “good times” and to contract in “bad times”. This managerial flexibility should change the exposure of the firm’s share price to the gold price in good times and bad times. The change is expected to yield an asymmetric effect which provides a higher exposure when the price of gold increases and a lower exposure when the price of gold decreases. The effect has option-like characteristics as the manager can exercise the real option to expand in good times and to contract in bad times.

To analyze the question whether managers of gold mining firms optimally exploit the real optionality of a gold mine we need to identify periods of increasing gold prices in which the exposure should increase. We define a dummy variable that is equal to one if gold price changes are positive and zero otherwise and augment the model in 3 by an interaction term comprised of gold returns and the dummy variable. We also vary the definition of the dummy variable to assess the sensitivity of the results with respect to the definition of the dummy variable. The augmented model can be written as follows

$$R_i = \alpha_i + \beta_{M,i}R_M + \beta_{G,i}R_G + \beta_{G,i}^*R_G D_{G,+} + \epsilon_i \quad (4)$$

with R_i , R_M and R_G being the returns of a gold mining firm i , the market return and the return of gold, respectively. The term $R_G D_{G,+}$ is an interaction term comprised of the return of gold and a positive gold return dummy variable. The error term is given by ϵ_i . The parameters to estimate are α_i , $\beta_{M,i}$, $\beta_{G,i}$ and $\beta_{G,i}^*$. The model estimates are used to test whether β_G^* is larger zero indicating an increased exposure to the price of gold in regimes of increasing gold prices. The formal presentation of this hypothesis is $H_0 : \beta_{G,i}^* > 0$.

Table 4 presents the coefficient estimates based on weekly returns for each of the 41 gold mining firms in the sample and shows that the exposure to gold price changes is larger in regimes of increasing gold prices. The average gold beta is 0.57 across all firms and increases by 0.21 to an average exposure of 0.78 in regimes of increasing gold prices. A comparison of the results with the symmetric model given by equation 3 shows that the symmetric model gold beta estimate of 0.67 is an average over all regimes and thus lies within the 0.57 and the 0.78 estimates obtained with the augmented model given by equation 4.

*** Insert Table 4 about here ***

The results are consistent with managers exploiting the real option characteristics of gold mining operations, i.e. gold mining firms exhibit an increased exposure to the gold price if gold prices increase relative to periods in which the price of gold does not change or decrease.⁷

3.2 Specification Issues

So far we have only estimated contemporaneous relationships and not considered dynamic (lagged) influences. However, it is possible that information incorporated in gold returns needs time to be processed by investors and thus time to be embedded in the share prices

⁷Similarly, if gold returns are negative, the average gold beta decreases by 0.20 from 0.77 to 0.57.

of gold mining firms. We included up to five lags of both the dependent variable and the independent variables for all return frequencies and find that lags do not influence the results of this study in a statistically or economically significant way. For example, lagged gold returns are not statistically or economically significant and the inclusion of lagged gold returns does not change the coefficient estimates of the contemporaneous effects or the fit of the model.

Another extension of the above analysis is to explicitly model the time-variation of firms' gold exposure. We use a 52-week moving window to estimate the gold beta for each firm and then use the cross-sectional average of gold betas at each point t to generate a time-series of cross-sectional gold betas. Figure 3 (top panel) presents the estimated time-series and illustrates that there is significant variation through time. The dynamic gold betas vary between values of zero (in the early and late 1980s), values clearly above one predominantly in the 1990s and values close to one in the 2000s. The period between 2000 and 2010 can be characterized by less volatile gold betas. The lower panel of the Figure presents the relationship of the time-varying gold beta and the price of gold via a scatter plot with connected points that additionally illustrate the dynamics through time. The scatter plot shows a significant variation of gold betas for gold prices around 400 US dollar and significantly less variation of gold betas for gold prices above 500 US dollar. Interestingly, the gold beta averages do not display a clear positive relationship with the price of gold consistent with real optionality over the whole range of gold prices but only for specific intervals, e.g. for prices between 500 and 700 and between 900 and 1,200.

*** Insert Figure 3 about here ***

The time-varying beta estimates are also consistent with the findings reported in other studies. Both Blose and Shieh (1995) for the period 1981 - 1990 and Tufano (1998) for the period 1990 - 1994 find gold betas around 1.5. Figure 3 illustrates that our estimates are similar despite the different sample of firms.

We have also examined the robustness of the asymmetric effect by changing the definition of the dummy variable. For example, instead of examining only positive and negative gold return regimes, we also estimated changes in gold betas for gold returns above or below a certain threshold. Not surprisingly, the changes in the beta are smaller for more extreme returns but still consistent with real optionality. Moreover, we analyzed low gold return and high gold return volatility regimes with different thresholds. The estimation results show that high volatility regimes are associated with smaller gold betas whilst low volatility regimes are associated with greater gold betas. Finally, we also examined specific periods such as the gold bull market between 2003 and 2010. The gold betas in that period decreased on average. One explanation for this finding is that gold mining firms first increased their production but significantly decreased it the longer the bull market lasted.

3.3 Cross-sectional Analysis - Managerial flexibility, Hedging and Diversification

This section briefly describes the results of an analysis of the 2010 or 2011 annual reports of all gold mining firms in the sample. We searched for information regarding managerial flexibility, hedging activities and the diversification of the firm. Around half of the firms (23 out of 41) address managerial flexibility and mention expansion options. Around a quarter of the firms (12 out of 41) report either current hedging activities or hedging in the past. Finally, 29 out of the 41 firms state that they do not only explore gold mines but are also engaged in the production of other commodities like silver, copper and uranium. A regression of the estimated betas on three binary indicator variables assuming a value of one if the firm is hedged, reports the use of managerial flexibility and is diversified yields only weak results both statistically and economically. It is not surprising that we find only weak evidence for a significant relationship between the variables and the estimated gold beta because the information about hedging, managerial flexibility and diversification is based on the 2010 and 2011 annual reports but the estimated beta is

based on a large number of observations prior to 2010. Unfortunately, it is not possible to compile information for each year from 1980 until 2011 for all companies (e.g. see Coleman (2010) for a study of two gold mining firms and an explicit focus on hedging). In this case betas for each year could be related to the hedged exposure in the same year. Finally, accounting for the size of the firm by including market capitalization tends to increase beta but this effect is driven by the largest mining companies in the sample (Anglogold Ashanti, Eldorado and Newcrest) which display higher than average gold betas.

4 Summary and Concluding Remarks

This study analyzed 41 Australian gold mining firms and its exposure to the price of gold and the Australian stock market for a 30-year period at different sampling frequencies and sub-periods characterized by bull and bear market conditions in gold. The study finds that the weekly sampled share price returns of the gold mining firms move by 0.7% for each 1% change in the price of gold denominated in US dollar. The gold beta or gold price exposure is larger for monthly returns and indicates that share prices move by 1.1% for each 1% change in the price of gold. The sensitivity to the gold price denominated in Australian dollars is significantly lower at 0.4% for weekly data and 0.75% for monthly data. The empirical analysis also indicates that managers exploit the real optionality embedded in the value of gold mines evidenced by the asymmetric gold exposure in bull and bear markets. Since hedging and diversification of gold mining firms do not imply an asymmetric exposure the real optionality can be distinguished from hedging and diversification.

The average sensitivity of Australian gold mining firms to the price of gold is close to one and thus similar to the exposure provided by bullion or exchange-traded funds on gold which became popular alternatives to investments in gold mining shares in the same period. The relatively low sensitivity can be explained with gold price hedging and the fact that many gold mining companies are diversified as they do not only produce gold but also silver and other metals.

Future research could examine alternative approaches to analyze real options embedded in gold mining projects. One potential approach could be to focus on the stock market listing and delisting of gold mining companies.

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Table 1: Descriptive statistics - Weekly returns

	mean	std. dev.	skewness	kurtosis	min	max	JB stat	p-value	LB stat	p-value	Nobs
ALKANE RESOURCES	-0.0001	0.1219	0.1443	10.0336	-0.7472	0.6931	2475.65	0.00	32.01	0.00	1199
ALLIANCE RESOURCES	-0.0013	0.1282	0.1253	5.1340	-0.4626	0.5108	162.56	0.00	13.86	0.02	845
ALLIED GOLD MINING CDI.	0.0039	0.0962	0.1677	5.0046	-0.3552	0.4140	63.34	0.00	6.04	0.30	368
AMPELLA MINING	0.0172	0.1395	-0.0868	4.4963	-0.4700	0.4700	16.64	0.00	8.24	0.14	176
ANGLOGOLD ASHANTI CDI.	0.0002	0.0534	-0.0061	4.8731	-0.2453	0.2280	84.79	0.00	14.72	0.01	580
AZUMAH RESOURCES	0.0039	0.1106	-0.2086	3.6000	-0.3365	0.2877	5.76	0.06	3.34	0.65	259
BEADELL RESOURCES	0.0041	0.1679	-0.2566	9.4786	-0.8873	0.6190	299.17	0.00	10.12	0.07	170
CGA MINING	0.0020	0.1334	-0.0034	11.7517	-0.6931	0.7621	3657.31	0.00	36.91	0.00	1146
CHALICE GOLD MINES	0.0042	0.0999	0.1889	4.1931	-0.3254	0.4055	16.18	0.00	21.20	0.00	248
ELDORADO GOLD CDI.	0.0048	0.0494	0.1275	2.9235	-0.0940	0.1392	0.16	0.92	7.54	0.18	55
EVOLUTION MINING	0.0014	0.1145	1.2638	8.7278	-0.3567	0.6168	712.08	0.00	32.20	0.00	436
FOCUS MINERALS	-0.0023	0.1328	0.3388	5.4762	-0.4055	0.6931	147.46	0.00	30.26	0.00	537
GOLD ONE INTERNATIONAL	-0.0069	0.1326	0.6571	12.5044	-0.5894	0.9555	1975.48	0.00	21.32	0.00	515
GOLDEN WEST RESOURCES	0.0036	0.1197	1.0554	7.0746	-0.3102	0.6221	274.63	0.00	3.11	0.68	313
GREENLAND MRLS.& ENERGY	0.0062	0.1769	5.1138	53.2323	-0.3629	1.8608	25840.87	0.00	1.50	0.91	236
IMX RESOURCES	0.0029	0.0895	0.3808	7.0617	-0.5543	0.4733	556.45	0.00	3.99	0.55	782
INTREPID MINES	0.0011	0.1060	-0.1779	7.7300	-0.5390	0.4590	537.17	0.00	8.45	0.13	573
KENTOR GOLD	-0.0026	0.1391	0.0310	6.6343	-0.6931	0.5596	165.70	0.00	9.48	0.09	301
KINGSGATE CONSOLIDATED	0.0047	0.0858	0.4582	8.2799	-0.3651	0.5581	935.70	0.00	4.54	0.47	782
KINGSROSE MINING	0.0088	0.1221	-0.2456	5.3020	-0.4055	0.4055	36.71	0.00	2.97	0.71	159
MARENGO MINING	0.0021	0.1134	0.2511	7.0500	-0.6286	0.4626	257.46	0.00	10.65	0.06	371
MEDUSA MINING	0.0086	0.0734	0.1080	4.2545	-0.2433	0.3102	24.71	0.00	2.84	0.73	366
MORNING STAR GOLD NL	0.0001	0.1449	-0.8026	18.3363	-1.4271	0.8473	7737.73	0.00	29.94	0.00	781
NEWCREST MINING	0.0016	0.0608	-0.0465	5.7747	-0.3804	0.2771	377.02	0.00	8.02	0.16	1174
NOBLE MINERAL RESOURCES	0.0038	0.1544	1.8588	14.8859	-0.5306	0.8575	840.09	0.00	4.86	0.43	130
NORTON GOLD FIELDS	0.0004	0.1209	0.3266	3.8655	-0.3677	0.4796	13.47	0.00	10.21	0.07	275
OCEANAGOLD CDI.	-0.0004	0.1185	-0.3269	5.3697	-0.4463	0.3950	46.33	0.00	8.71	0.12	184
PENINSULA ENERGY	0.0000	0.1730	0.0644	7.6622	-0.6931	0.6931	783.99	0.00	48.04	0.00	865
PERSEUS MINING	0.0093	0.0978	0.2678	5.5293	-0.3637	0.4329	91.07	0.00	7.77	0.17	327
RAMELIUS RESOURCES	0.0052	0.1174	2.9273	27.1985	-0.2787	1.1550	10434.01	0.00	5.23	0.39	404
RED 5	0.0003	0.1171	0.2544	4.5949	-0.4055	0.4463	86.53	0.00	15.72	0.01	741
REGIS RESOURCES	0.0006	0.1383	-0.8380	27.6650	-1.3863	1.0986	28801.43	0.00	3.02	0.70	1131
RESOLUTE MINING	0.0002	0.0776	0.1178	6.1162	-0.4418	0.3512	466.33	0.00	3.66	0.60	1146
SARACEN MINERAL HDG.	0.0042	0.1621	3.8515	63.0453	-1.0986	2.3026	119410.69	0.00	6.20	0.29	782
SIHAYO GOLD	-0.0007	0.1380	0.3205	5.7380	-0.6931	0.6931	257.65	0.00	24.52	0.00	782
SILVER LAKE RESOURCES	0.0110	0.1032	0.5582	3.9940	-0.2624	0.3677	15.17	0.00	5.08	0.41	163
SOUTH BOULDER MINES	0.0059	0.1460	1.1724	9.9199	-0.5108	1.0116	834.12	0.00	5.90	0.32	375
ST BARBARA	0.0000	0.1043	0.3469	8.4638	-0.6931	0.6931	2043.80	0.00	14.90	0.01	1617
TALISMAN MINING	0.0057	0.1485	0.6947	5.9674	-0.4700	0.7282	118.54	0.00	3.25	0.66	265
TANAMI GOLD	-0.0012	0.1076	0.6678	8.2678	-0.5108	0.7563	1406.54	0.00	15.12	0.01	1143
TROY RESOURCES	0.0038	0.0816	0.1066	5.9118	-0.4329	0.3909	277.73	0.00	14.38	0.01	782
Australian stock market index	0.0015	0.0248	-1.8608	29.1070	-0.3503	0.1193	46854.22	0.00	17.67	0.00	1617
Gold (USD)	0.0006	0.0254	0.3990	9.4933	-0.1238	0.2133	2883.60	0.00	10.36	0.07	1617
Gold (AUD)	0.0007	0.0253	0.6309	9.4289	-0.1578	0.1565	2891.96	0.00	4.08	0.54	1617

The JB (Jarque-Bera) statistics and the corresponding p-values indicate that the returns are not normally distributed.

The LB (Ljung Box) statistics and the corresponding p-values indicate that the majority of returns do not exhibit significant serial correlation.

Table 2: **Average market betas and gold betas**

The Table presents the coefficient estimates (averaged across all firms) of five different model specifications for daily, weekly and monthly returns. The estimates show that the market beta, the gold beta and the model fit increase from daily to monthly returns.

	Australian stock index	Gold in USD	Gold in AUD	R squared
daily				
Spec.1	0.71***	0.20		0.03
Spec.2	0.79***		0.33	0.04
Spec.3	0.73***			0.02
Spec.4		0.25		0.01
Spec.5			0.21	0.01
weekly				
Spec.1	0.91***	0.67*		0.09
Spec.2	1.12***		0.61	0.09
Spec.3	1.00***			0.05
Spec.4		0.77*		0.05
Spec.5			0.40	0.03
monthly				
Spec.1	1.10*	1.13*		0.18
Spec.2	1.43***		0.65	0.13
Spec.3	1.16*			0.08
Spec.4		1.19*		0.11
Spec.5			0.13	0.05

*, ** and *** denote statistical significance at the 10%, 5% and 1% level, respectively.

Figure 1: **Price of gold in USD and in AUD**

The solid line presents the price of gold in US dollar and the dashed (blue) line shows the price of gold in Australian dollar.

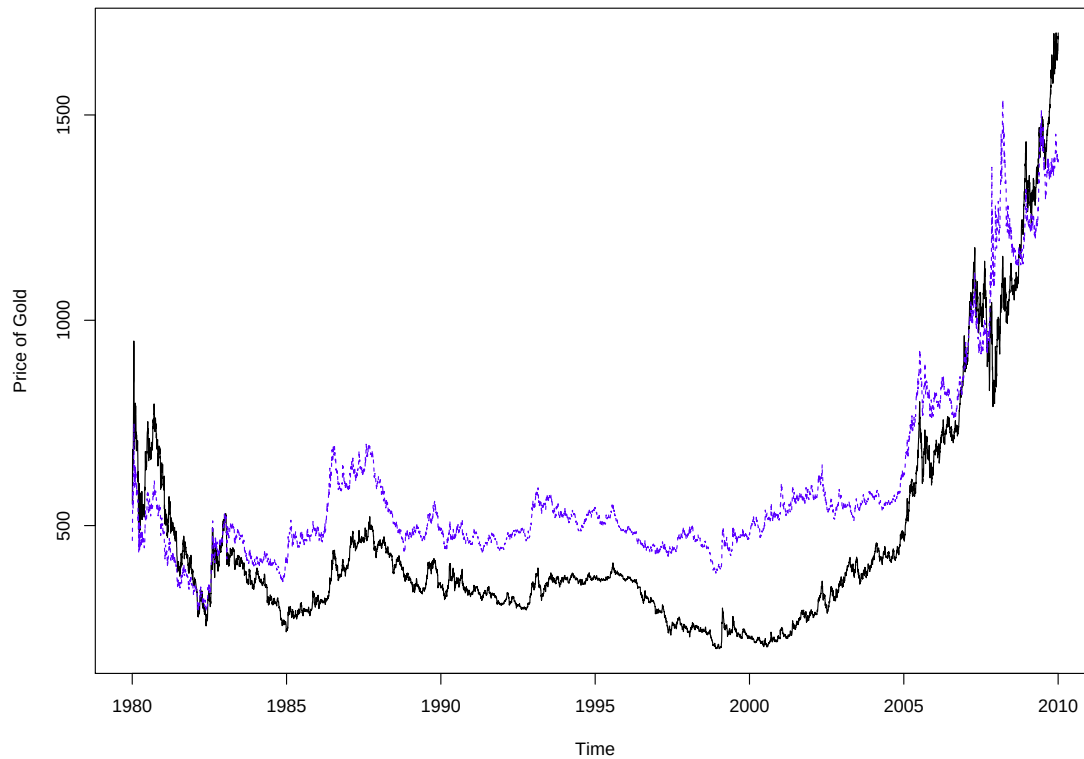


Table 3: Market beta and gold beta

This Table presents the estimation results for weekly returns based on the model $R_i = \alpha_i + \beta_{M,i}R_M + \beta_{G,i}R_G + \epsilon_i$

	Market	Gold	R squared	Nobs
ALKANE RESOURCES	0.785 ***	0.722 *** #	0.04	1199
ALLIANCE RESOURCES	1.354 ***	0.321 * ###	0.05	845
ALLIED GOLD MINING CDI.	1.223 ***	1.043 ***	0.21	368
AMPELLA MINING	0.743 **	0.145 ##	0.03	176
ANGLOGOLD ASHANTI CDI.	0.373 ***	0.854 *** #	0.19	580
AZUMAH RESOURCES	1.329 ***	0.623 *** #	0.16	259
BEADELL RESOURCES	0.517	0.251 #	0.01	170
CGA MINING	0.616 ***	0.276 ###	0.01	1146
CHALICE GOLD MINES	0.436 *	0.303 ###	0.03	248
ELDORADO GOLD CDI.	-0.074	0.927 ***	0.17	55
EVOLUTION MINING	1.151 ***	1.024 ***	0.12	436
FOCUS MINERALS	0.788 ***	0.403 * ###	0.02	537
GOLD ONE INTERNATIONAL	0.570 **	1.293 ***	0.08	515
GOLDEN WEST RESOURCES	1.051 ***	0.593 *** #	0.08	313
GREENLAND MRLS.& ENERGY	0.875 **	0.324 #	0.02	236
IMX RESOURCES	1.106 ***	0.409 *** ###	0.08	782
INTREPID MINES	1.067 ***	0.814 ***	0.09	573
KENTOR GOLD	0.250	0.918 ***	0.04	301
KINGSGATE CONSOLIDATED	1.127 ***	0.804 ***	0.13	782
KINGSROSE MINING	0.582 *	0.143 ###	0.02	159
MARENGO MINING	1.337 ***	0.617 *** #	0.11	371
MEDUSA MINING	1.003 ***	0.616 *** ###	0.19	366
MORNING STAR GOLD NL	0.996 ***	0.553 ** ##	0.03	781
NEWCREST MINING	0.738 ***	1.296 *** ###	0.28	1174
NOBLE MINERAL RESOURCES	1.043 **	0.142 ##	0.05	130
NORTON GOLD FIELDS	0.918 ***	0.901 ***	0.11	275
OCEANAGOLD CDI.	1.141 ***	1.013 ***	0.17	184
PENINSULA ENERGY	1.145 ***	0.599 **	0.03	865
PERSEUS MINING	1.096 ***	0.676 *** #	0.14	327
RAMELIUS RESOURCES	1.175 ***	0.839 ***	0.1	404
RED 5	0.951 ***	1.001 ***	0.08	741
REGIS RESOURCES	0.577 ***	0.785 ***	0.02	1131
RESOLUTE MINING	0.934 ***	1.470 *** ###	0.23	1146
SARACEN MINERAL HDG.	1.112 ***	0.423 * ##	0.03	782
SIHAYO GOLD	1.117 ***	0.541 *** ##	0.04	782
SILVER LAKE RESOURCES	1.187 ***	1.167 ***	0.28	163
SOUTH BOULDER MINES	1.496 ***	0.431 ##	0.07	375
ST BARBARA	0.843 ***	0.749 *** ##	0.09	1617
TALISMAN MINING	1.379 ***	-0.012 ###	0.06	265
TANAMI GOLD	0.555 ***	0.841 ***	0.04	1143
TROY RESOURCES	0.682 ***	0.721 *** ##	0.08	782

*, ** and *** denote statistical significance at the 10%, 5% and 1% level, respectively.

#, ## and ### denote statistical significance at the 10%, 5% and 1% level, respectively, based on the null hypothesis that the coefficient for gold is equal to one.

Table 4: Market beta and gold beta with asymmetry

This Table presents the estimation results for weekly returns obtained with the model $R_i = \alpha_i + \beta_{M,i}R_M + \beta_{G,i}R_G + \beta_{G^*,i}R_G D_{G,+} + \epsilon_i$ where $D_{G,+}$ is a dummy variable that is equal to one if gold returns are positive and zero otherwise.

	Market	Gold	Gold +	R squared	Nobs
ALKANE RESOURCES	0.776 ***	0.838 ***	-0.224	0.04	1199
ALLIANCE RESOURCES	1.328 ***	0.581 *	-0.484	0.05	845
ALLIED GOLD MINING CDI.	1.253 ***	0.823 ***	0.438	0.21	368
AMPELLA MINING	0.813 **	-0.492	1.231	0.04	176
ANGLOGOLD ASHANTI CDI.	0.398 ***	0.627 ***	0.440 *	0.2	580
AZUMAH RESOURCES	1.259 ***	1.225 ***	-1.207 *	0.17	259
BEADELL RESOURCES	0.531	0.115	0.265	0.01	170
CGA MINING	0.627 ***	0.136	0.269	0.01	1146
CHALICE GOLD MINES	0.393 *	0.680 *	-0.755	0.03	248
ELDORADO GOLD CDI.	-0.086	1.106 *	-0.343	0.17	55
EVOLUTION MINING	1.246 ***	0.275	1.491 **	0.14	436
FOCUS MINERALS	0.804 ***	0.265	0.269	0.02	537
GOLD ONE INTERNATIONAL	0.626 **	0.830 **	0.910	0.08	515
GOLDEN WEST RESOURCES	1.029 ***	0.769 *	-0.347	0.08	313
GREENLAND MRLS.& ENERGY	0.830 **	0.755	-0.832	0.03	236
IMX RESOURCES	1.086 ***	0.594 **	-0.346	0.08	782
INTREPID MINES	1.154 ***	0.025	1.521 ***	0.1	573
KENTOR GOLD	0.192	1.412 ***	-0.976	0.05	301
KINGSGATE CONSOLIDATED	1.123 ***	0.848 ***	-0.082	0.13	782
KINGSROSE MINING	0.593 *	0.038	0.208	0.03	159
MARENGO MINING	1.288 ***	0.986 ***	-0.736	0.12	371
MEDUSA MINING	1.071 ***	0.100	1.029 ***	0.2	366
MORNING STAR GOLD NL	1.040 ***	0.127	0.797	0.03	781
NEWCREST MINING	0.761 ***	1.006 ***	0.556 ***	0.28	1174
NOBLE MINERAL RESOURCES	1.021 **	0.281	-0.277	0.05	130
NORTON GOLD FIELDS	0.982 ***	0.344	1.110	0.12	275
OCEANAGOLD CDI.	1.139 ***	1.031 **	-0.035	0.17	184
PENINSULA ENERGY	1.194 ***	0.077	0.974	0.03	865
PERSEUS MINING	1.123 ***	0.470	0.405	0.14	327
RAMELIUS RESOURCES	1.228 ***	0.440	0.788	0.11	404
RED 5	0.898 ***	1.504 ***	-0.942 *	0.08	741
REGIS RESOURCES	0.625 ***	0.186	1.144 **	0.03	1131
RESOLUTE MINING	0.936 ***	1.443 ***	0.053	0.23	1146
SARACEN MINERAL HDG.	1.113 ***	0.414	0.018	0.03	782
SIHAYO GOLD	1.081 ***	0.885 **	-0.641	0.04	782
SILVER LAKE RESOURCES	1.261 ***	0.437	1.430 **	0.3	163
SOUTH BOULDER MINES	1.494 ***	0.449	-0.035	0.07	375
ST BARBARA	0.848 ***	0.654 ***	0.178	0.09	1617
TALISMAN MINING	1.394 ***	-0.138	0.252	0.06	265
TANAMI GOLD	0.596 ***	0.321	0.996 **	0.05	1143
TROY RESOURCES	0.681 ***	0.736 ***	-0.028	0.08	782

*, ** and *** denote statistical significance at the 10%, 5% and 1% level, respectively.

Figure 2: **Histogram of market betas and gold betas**

The histogram shows the distribution of market betas (left panel) and gold betas (right panel) based on weekly returns for all 41 gold mining firms in the sample.

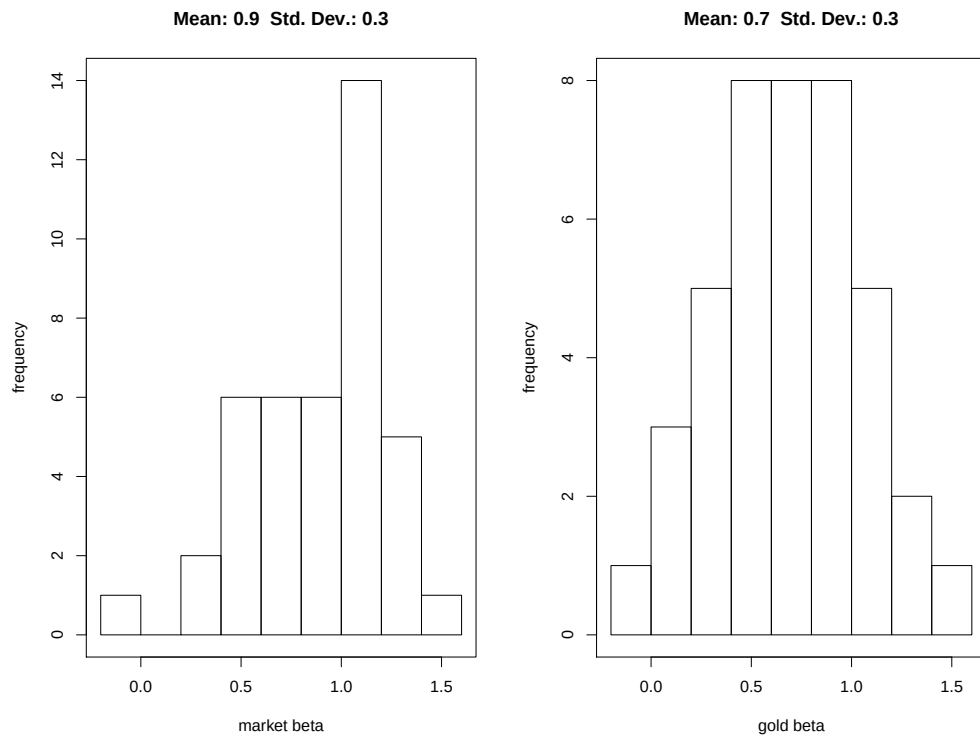


Figure 3: **Time-varying beta of gold**

This Figure presents the time-varying beta of gold calculated as the cross-sectional average of individual betas estimated using weekly returns and a 52-week moving window (top graph) and the relationship of the time-varying beta with the price of gold (bottom graph). The graphs illustrates significant variation of the gold betas through time and for different gold price regimes.

